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Digital Forensic Readiness Information System For EJBCA Digital Signature Web Server

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***Abstract*— As the nature of the website, the EJBCA digital signatures may have vulnerabilities. The list of web-based vulnerabilities can be found in OWASP's Top 10 2021. Anticipating the attack with an effective and efficient forensics application is necessary. The concept of digital forensic readiness can be applied as a pre-incident plan with a digital forensic lifecycle pipeline to establish an efficient forensic process. Managing digital evidence in the pre-incident plan includes data collection, examination, analysis, and findings report. Based on this concept, we implemented it in designing an information system that carries out the entire flow, provides attack evidence collection, visualization of attack statistics in executive summary, mitigation recommendation, and forensic report generation in a physical form when needed. This research offers an information system that can help the digital forensic process and maintain the integrity of the EJBCA digital signature server web.**

Keywords—Digital forensic readiness, engineering design process, digital forensics investigation lifecycle

I. INTRODUCTION

A digital signature is one of the mechanisms of the parties involved to provide certainty regarding the authenticity and approval of documents by the signer. Digital signatures as a commitment binding mechanism utilize the sign and verify algorithm [1]. Digital signatures are necessary because they have a property that is non-repudiation. Non-repudiation means Assurance that both the sender and the recipient of the information are given proof of delivery and the sender's identity, so neither can subsequently deny processing the information [2]. One application of digital signatures used in various important instances is the Enterprise Java Beans Certificate Authority (EJBCA) system with the Public Key Infrastructure (PKI) architecture [3]. As a web-based digital signature system, EJBCA could potentially be exposed to attacks against the web as described in the 2021 OWASP Top 10 survey [4].

Various harmful exploits on EJBCA as digital signatures without addressing vulnerabilities can lead to losses. Based on the taxonomic survey of attacks that occur on digital signatures conducted by Hernandez et al., the results of damaging exploitation of digital signatures are based on the attacker's goals which include [5]:

- Tricking the signer into signing a document that the attacker wants, but is different from the document that was supposed to be signed;
- Unauthorized use of signature creation data (SCD);
- Replace stored signature information at the attacker's discretion;
- Attribute a signed document to a different user than the actual signer;
- Make Data To Be Verified (DTBV) to display selected content; and
- Make the signature validity verification end with the opposite result.

These losses can lead to a decrease in integrity and reputation if not addressed, so it is necessary to design a digital forensic readiness information system for the EJBCA digital signature web server.

II. RESEARCH METHODOLOGY

The methodology that will be used in this research is the engineering design process (EDP). The use of EDP was chosen as a guide because it follows a system's engineering needs. In this research, the EDP method is used to guide the design of a digital forensic readiness information system. Based on this methodology, the questions that form the basis for this research can be obtained, the alternative concepts offered, testing the engineering implementation of the concept, and evaluating the performance of the engineering results based on testing.

A. Research Questions

In this research, there are two research questions that will be addressed:

1. What are the specific steps required to implement digital forensics readiness for EJBCA web server?
2. What digital forensics readiness' features that can be implemented in EJBCA digital signature web server to help report the attack?

The selection of these two questions is based on the EJBCA structure which is a web-based PKI [3] that could potentially be exposed to attacks against the web. These types of potential attacks are described in the 2021 OWASP Top 10 survey[4]. Digital forensic readiness is defined as a pre-incident plan for the digital forensic investigation lifecycle process [6] that can be used to anticipate future attacks, in this case, against the EJBCA web component.

B. Generate Alternative Concepts

Based on OWASP's top 10 (2021) and the concept of the EJBCA web server, an evaluation is needed in system transaction records, the status of data transfer on the system network, and the status of the module and encryption on the system. This evaluation will be considered in the digital forensic readiness information system design.

After a literature review, we found three alternative concepts that can be implemented. The first alternative concept is using EJBCA logs as a piece of evidence. It contains transaction records in the form of input and output data of EJBCA system users as long as the system is active [7]. The second alternative is implementing Snort intrusion detection system (IDS) tools. Snort will be used as the basis

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for digital evidence regarding the data transfer status on the system network [8], internet protocol (IP) address data and traffic network on the EJBCA system. The last alternative solution is to collect server logs. Collecting proof of module status and encryption on the system was carried out through server logs accessed through the Ubuntu terminal [9], [10].

Three alternative concepts of digital forensic readiness solutions for EJBCA digital signatures have differences in the format and content of each record. The difference between the three alternative concept solutions in recording can occur because each alternative solution has a different scope for monitoring digital data. The alternative EJBCA log solution includes user input and output transaction data on system applications. The alternative solution using Snort IDS covers recording network aspects on the EJBCA system. In contrast, the alternative solution for collecting evidence through server logs accessed through the Ubuntu terminal includes data recorded by the operating system. We limit access to avoid disrupting the operating system.

C. *Evaluate and Select a Concepts*

The three alternative solution concepts that have been determined will be confirmed and a solution design will be made which will be implemented in the final research of this task. Each alternative concept has advantages and disadvantages based on evidence data monitored on the EJBCA system. The results of these advantages and disadvantages will be considered to find the right solution to this problem. Evaluation of all alternative concepts is arranged in the form of a comparison table shown in TABLE I.

TABLE I. DIGITAL FORENSIC EVIDENCE PARAMETERS LISTED ON EACH ALTERNATIVE SOLUTION

Evidence Parameters Digital Forensics	Alternative Solutions		
	EJBCA Application Logs	Log IDS Tools Snort	Log Server Via Terminal
Timestamp	V	V	V
Log level	V	X	X
Event Module	V	X	X
Server Service Thread Pool Port	V	X	X
Event Description	V	V	V
Input/Output Packages (IO)	V	V	V
Date and time of event	V	V	V
SID	X	V	X
Attack alert messages	X	V	X
Priority	X	V	X
Internet Protocol (IP)	V	V	V
IP address	V	V	V
Port	X	V	X
TLS version	X	X	V
Handshake Status	X	X	V
Path Certificate	X	X	V
Modul name	V	V	V
Modul version status	X	X	V

D. *Details of the Design*

The design of the digital forensic readiness information system for the EJBCA digital signature's web server will be implemented with the basic reference of the digital forensic lifecycle process flow that is included in the digital forensic readiness proposed by Mouhtaropoulos et al [6]. An overview of the digital forensic lifecycle process from beginning to end is compiled in the form of a swimlane diagram model which is illustrated in Fig. *1*.

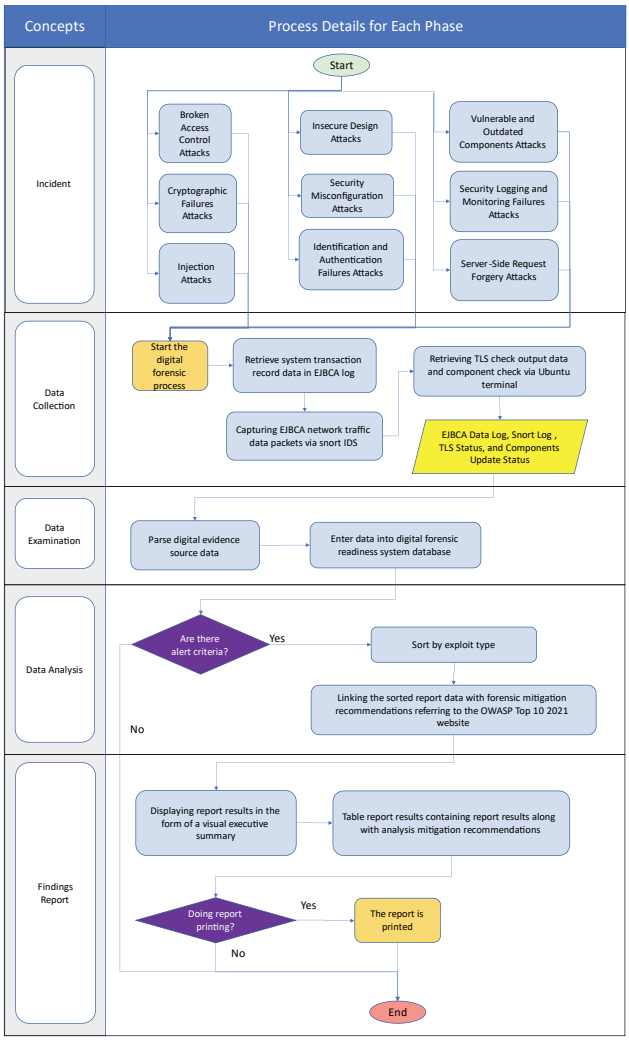

Fig. 1. Digital Forensic Life Cycle Process Flow Managed in Digital Forensic Readiness

Based on the swimline diagram above, digital forensic lifecycle managed in digital forensic readiness concept is applied with specific steps for each phase. Data collection starts when an incident has occured. The system then retrives system transaction record data in EJBCA log, captures data packets via snort, and retrives TLS check output data and component check via Ubuntu terminal. All of these information will then be parsed during data examination. These parsed data will be entered into digital forensic readiness system database. Data analysis phase will sort the content of said database according to their exploit type and link each one of them to forensic mitigation recommendations referring to the OWASP Top 10 2021 website. Findings report phase presents the information from previous stage in the form

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of a visual executive summary, complete with table report results containing details and linked recommendations.

Based on the description presented, it can be described that the design of the digital forensic readiness information system will manage the four stages of the digital forensic life cycle, namely: data collection, data examination, analysis, and report findings. At the data collection stage, the processing system collects data from digital evidence sources that will be used with digital evidence sources:

- EJBCA application logs;
- IDS snort logs; and
- Server logs via terminal.

Then, at the data examination stage, the decomposition of data sources that can be used as digital evidence is carried out which is then entered into the basic data system using technology:

- Python parser script;
- Flask web framework for python APIs; and
- Firebase database.

After the data is entered into the database, data analysis is carried out based on the match between alerts and digital evidence source data implemented in the Python API with the flask web framework. After the data is analyzed, the data processing results are used as findings report and displayed in the form of an executive summary. It can be printed as physical evidence by using technology:

- Firebase deployment platform for web interface development scope; and
- Flutter language as web interface frontend implementation.

E. *Manufacture and Testing*

The design of a digital forensic readiness information system for EJBCA digital signatures is implemented as data log system with a website interface. Based on the details of the information system design, the specifications for each component of the information system design are determined as follows.

- EJBCA server application log version 7.4.3
- IDS version 2.9.7 snort logs;
- Ubuntu version 20.04 terminal output log;
- Python language parser script version 3.8.10;
- Flask web framework version 2.1.3 for the python API;
- Firebase database that is NoSQL cloud database;
- The firebase deployment platform is cloud-hosted for web interface development scopes; and
- Flutter language version 3.0.5 as web interface frontend implementation

The components of the digital forensic readiness information system design with these specifications will be implemented with a description in the form of an illustration in Fig. 2. After the system is implemented, testing is carried out on the performance of the information system. The test is carried out in the form of a simulated attack on the EJBCA digital signature web server which is illustrated in Fig. 3.

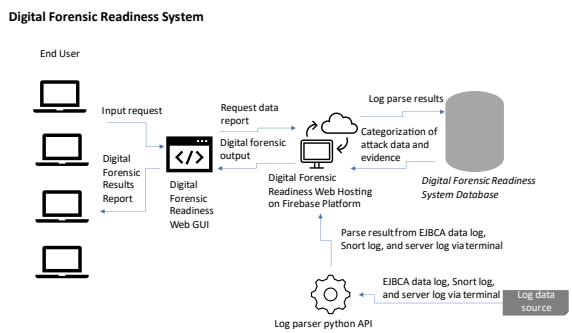

Fig. 2. Digital Forensic Readiness Information System Design

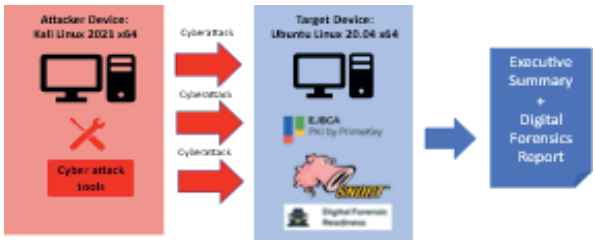

Fig. 3. Digital Forensic Readiness Implementation Testing Scenario

The implementation of the scenario is carried out by simulating the OWASP Top 10 attack in 2021 with the medium-level DVWA standard. The method of carrying out the attack can be carried out with a series of guidelines that can be seen on GitHub : "OWASP Top 10 2021 Attacks Sample Guide Based on DVWA Medium Level Standard" written by Lorddeepwhite [11].

F. *Evaluation of Performance*

Digital forensic readiness requires features that can fulfil the entirety of digital forensic lifecycle process. Said features include attack evidence collection, visualization of attack statistics, mitigation recommendation, and forensic report generation in a physical form. The performance of the digital forensic information system readiness was declared to have met the needs of the research question when it met the parameters used for testing the performance of the specified system implementation. Features that have met parameters should be able to fulfil the needs of digital forensic readiness implementation. Parameters stated in the code "EOP-XX" to facilitate mapping, as well as the use of proportion figures so that the results obtained can be achieved more precisely. The test parameters are shown in TABLE II.

TABLE II. PERFORMANCE TESTING PARAMETERS FOR DIGITAL FORENSIC READINESS SYSTEM IMPLEMENTATION

Parameter Code	Parameters	Types of Assessments	Description
EOP-01	Conformity of the digital forensic investigation lifecycle process flow concept with the applied digital forensic readiness system performance flow	Percentage: (number of stages of digital forensic flow carried out by the system / number of stages of digital forensic flow in the concept of digital forensic readiness) x100%	The percentage of conformity of the digital forensic investigation lifecycle process flow concept to the applied digital forensic readiness system performance flow will serve as proof of how to implement the concept into an information system design.

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Parameter Code	Parameters	Types of Assessments	Description
EOP-02	Comparison of the number of attacks with recorded evidence against the Top 10 OWASP attack types in 2021	Percentage: (number of detected attack types /10x100%	The percentage of the comparison between the number of attacks and the evidence recorded against the 10 types of OWASP Top 10 attacks in 2021 will serve as the level of gathering forensic evidence in the digital forensic readiness system
EOP-03	Comparison of the number of attacks shown in the executive summary against the Top 10 OWASP attacks in 2021	Percentage: (number of attack types displayed / 10)x 100%	The percentage of the comparison between the number of types of attacks displayed against the 10 types of OWASP Top 10 attacks in 2021 will serve as the level of availability of information in the executive summary section of the digital forensic readiness system
EOP-04	Availability of information on forensic mitigation recommendations against the 10 types of OWASP Top 10 attacks in 2021	Percentage: (number of mitigation recommendation types/10x100%	The percentage of comparison between the number of types of recommendations available on digital forensic readiness with the 10 types of OWASP Top 10 attacks in 2021 will work as the level of availability of recommendation mitigation information on digital forensic readiness systems
EOP-05	Availability of printable forensic evidence reports	Availability Status: "Available" or "Not available"	Status is intended to provide information on the availability of the report function on the digital forensics system's readiness to print evidence as a physical evidence report

III. RESEARCH RESULT

The test results of each attack sampled for performance assessment of digital forensic readiness information systems are shown from the match between the attack data recorded in the log source and that displayed on the system interface. For example, the match between the DVWA standard broken access control sample data and the medium level recorded in the EJBCA log in Fig. 4 to the display of the attack evidence record on the digital forensic readiness web interface in Fig. 5.

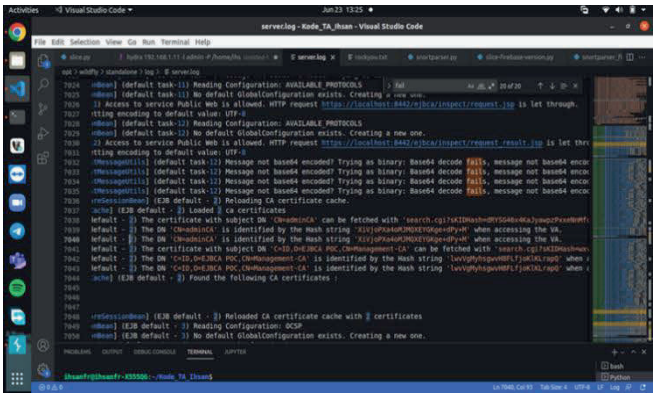

Fig. 4. EJBCA log displays broken access control attack records with medium-level DVWA standard file upload mechanism

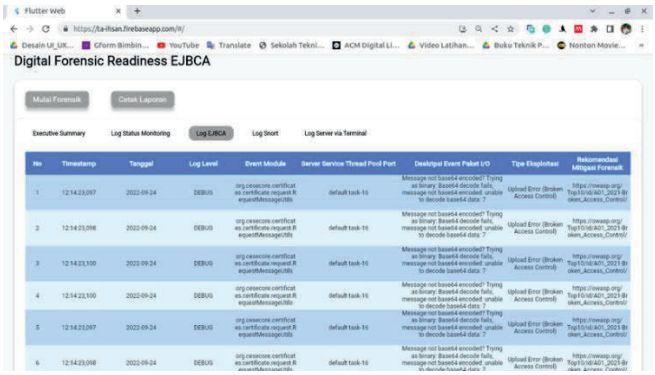

Fig. 5 Display on the digital forensics web interface readiness related to broken access control attack records with medium-level DVWA standard file upload mechanism

Each inspection result performed by the system is presented in a concise visual display on the executive interface. This is illustrated in Fig. 6.

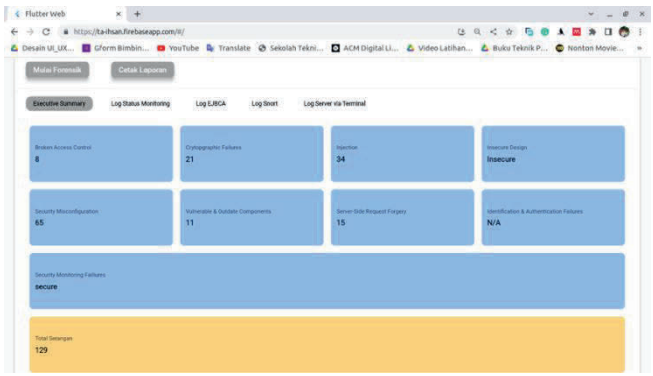

Fig. 6. Executive summary visual display showing digital forensic process results on digital forensic readiness web

The results of this research regarding the digital forensics readiness against predetermined test parameters are shown in TABLE III.

TABLE III. TEST RESULTS FOR DIGITAL FORENSIC READINESS IMPLEMENTATION

Parameter Code	Parameters	Assessment Results	Description
EOP-01	Conformity of the digital	100%	The percentage of 100% is obtained from

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Parameter Code	Parameters	Assessment Results	Description
	forensic investigation lifecycle process flow concept with the applied digital forensic readiness system performance flow		the number of stages of the digital forensic investigation lifecycle processed by the information system in carrying out forensics on each attack compared to the digital forensic investigation lifecycle process flow stated in the concept of digital forensic readiness
EOP-02	Comparison of the number of attacks with recorded evidence against the Top 10 OWASP attack types in 2021	80%	The percentage of 80% is obtained from the total number of types of attacks with recorded evidence of eight, while the total types of attacks declared by the OWASP Top 10 in 2021 are ten
EOP-03	Comparison of the number of attacks shown in the executive summary against the Top 10 OWASP attacks in 2021	80%	The executive summary has shown the varying number of attacks carried out against the 10 types of attacks on behalf of the OWASP Top 10 of 2021
EOP-04	Availability of information on forensic mitigation recommendations against the 10 types of OWASP Top 10 attacks in 2021	80%	The percentage of 80% obtained from the number of types of forensic mitigation recommendations provided is eight, while the total types of attacks declared by the OWASP Top 10 in 2021 is ten
EOP-05	Availability of printable forensic evidence reports	Available	The print function of the forensic evidence report can function in accordance with the function's purpose, which is to produce a printed report in PDF format containing attack evidence data, executive summary, and forensic mitigation recommendations

In the test of digital forensic readiness implementation, there are five parameters. EOP 01 describes the conformity of the digital forensic investigation lifecycle process flow concept with the applied digital forensic readiness system performance flow. EOP 02 describes the number of types of attacks that can be obtained as evidence. EOP 03 provides an executive summary. EOP 04 provides a reference for forensic mitigation recommendations related to detected attacks. EOP 05 provides printed forensic reports in physical form when needed.

The results of the 100% value obtained in the EOP-01 parameter testing related to the stages of the digital forensics investigation lifecycle can be applied to the performance of the digital forensic readiness information system design. 80% of the number of attacks is obtained from testing the EOP-02 parameter because two attacks cannot be detected by implementing a digital forensic readiness system. The value of 80% for EOP-03 parameter is because of two attacks that cannot be detected by implementing the digital forensic readiness information system design, so forensic mitigation recommendations are not shown for the two attacks. The assessment of 80% of the tests on the EOP-04 parameters regarding the completeness of the executive summary was due to the absence of information related to two types of attacks that could not be detected by the implementation of the digital forensic readiness information system design. The "available" assessment results in the EOP-05 parameter are indicated by the printout of physical evidence reports that can be generated when the report print function is activated on the digital forensic readiness system interface.

IV. CONCLUSION

Based on the entire process of carrying out this final project, conclusions can be drawn that answer the research questions of this final project. The conclusion covers the following number of points.

1. The specific steps required to implement a digital forensic readiness for EJBCA digital signatures are detailed process for each phase of digital forensic investigation's life cycle process flow (data collection, examination, analysis, and findings reports). Data collection starts by collecting various logs and store them into a database. Data examination parses digital evidence source data from said database. Data analysis sorts the parsed data and link them to a forensic mitigation recommendations based on the attack type. Findings report presents the result from previous analysis and linked recommendation in the form of visual executive summary and table containing the details.
2. The digital forensics readiness' features that can be implemented in EJBCA digital signature web server to help report the attack are attack evidence collection, visualization of attack statistics in executive summary, mitigation recommendation, and forensic report generation in a physical form. Attack evidence collection contained all relevant data to an attack pattern which is necessary to ease attack type analysis. Visualization of attack statistics in executive summary helps to show the number of attack occurance. Mitigation recommendations help the investigator to form an immediate plan of action. Forensic report generation in a physical form make a presentable output for the law enforcement when needed.

This research shows that the concept of digital forensic readiness can be implemented in the form of an information system. But further development can be provided in future work. This can be done by increasing the number of attacks and the level of complexity of attacks that need to be detected by the system. In addition, improvements to the report format provided by the system can help make it easier for users to understand the result of reports generated by the system.

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